

AIRLINE FUEL CONSUMPTION PREDICTION USING MACHINE LEARNING

Aljowharah Alrasheed | ML Foundations Capstone Report

1. Introduction

I chose the Airline Route Profitability dataset to learn how machine learning can be used in the aviation industry. The goal of this project was to predict whether a flight has high or low fuel consumption using flight and financial data.

2. Dataset Description

The dataset contains information about airline flights, including passengers, flight hours, revenue, costs, and route details. Missing values were handled using the median, and a new target variable called `Fuel_Consumption_Level` was created to classify flights as having high or low fuel consumption.

3. Models Used

I trained three machine learning models:

- Logistic Regression
- Random Forest
- K-Nearest Neighbors (KNN)

The models were evaluated using Accuracy, Precision, Recall, and F1-score. Logistic Regression achieved the best overall performance.

4. Neural Network

I built a neural network using TensorFlow and Keras with Dense and Dropout layers. The model was trained for 10 epochs and evaluated on the test set. Its performance was then compared with the machine learning models.

5. Results & Comparison

The best model achieved an Accuracy of approximately 97.9% and an F1-score of approximately 97.9%. The neural network also produced good results, but Logistic Regression performed slightly better and was more efficient. Therefore, it was selected as the final model.

I also developed an interactive Gradio interface that allows users to enter flight information and receive fuel consumption predictions.

6. What I Learned

Through this project, I learned how to prepare data, handle missing values, train and evaluate machine learning models, build a neural network, and create an interactive user interface. I also learned the importance of comparing multiple models to select the best one.